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# DYNAMIC 3D EROSION OF TERRAINS

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# Chapter 1

## Introduction

Virtual terrains appear in a wide variety of applications: from entertainment to physical simulation or science visualization, a lot of virtual scenes are set on a digital landscape. Despite the ubiquity of virtual terrains on 3D scenes, the complex and fractal structure of natural landscapes still brings multiple challenges for the field of synthetic terrain generation.

Most virtual terrains are represented using 2D heightfields which, despite being highly convenient when generating, storing and manipulating terrain models, have a major drawback: heightfields can only represent 2.5D models. Because each point in a 2D plane has a set elevation and every point between the ground and the given elevation is considered as solid (and the terrain interior), we can not represent features like caves, arches or overhangs, where not all points between the ground and the maximum elevation are solid. Additionally, these representations of terrains also fail to reproduce complex geometry commonly found in rocky terrains, like sharp ridges, loose blocks or steep cliffs.

Many of these complex structures that can not be easily captured by 2D heightfields, arise from different types of weathering events, like percolation and freeze-thaw cycles. To address this limitations on virtual terrain models, our goal in this thesis is to study the mechanical weathering of rocky terrains to enhance models given from 2D heightfields and add the complex features that these fields fail to capture.

Terrain enhancement through weathering simulation not only improves the realism and quality of the models, adding the aforementioned complex features, but it can also be applied dynamically to simulate the changes a terrain goes through over multiple years.

This project is built on a previous project on this same topic [1], using the same algorithm as a base, we worked on addressing some limitations of the previous project.

### 1.1 Contributions

As mentioned before, this project is heavily based on a previous one, and our proposed algorithm is built on the previous one. Despite that, we add multiple improvements to the base algorithm to increase the quality of the results and its efficiency.

The main point we wanted to address of the previous project was the lack of a physically based approach

to the algorithm, as the proposed algorithm made a lot of simplifications and ignored multiple physical interactions which made the method less accurate and realistic. However, during this thesis we also found other areas that could be improved and expanded to improve both the quality and the efficiency of the algorithm.

The limitations of the previous project and our improvements will be discussed at length during these thesis, but our main contributions can be summarized in three large blocks: improvements to the decomposition of the terrain into cells, improvements in the efficiency of the algorithm and improvement on the quality of the algorithm by extending the simulation to a physically based one.

## 1.2 Set-up and tools

Another difference between the project we base this thesis on, albeit less significant, is the tools used. In the project that presented the algorithm we are extending, a blender plugging was implemented to show the method, however, here we have implemented our erosion model using C++ and we showed the results in a QT application using OpenGL for rendering.

To make the application, we reused the code provided by an article on terrain descriptors [2], as it already implemented many basic functionalities to work with virtual terrains. We then modified and expanded the application implementing our algorithm for mechanical weathering as well as implementing other functionalities to aid in the visualization and debugging of our method. Some of the changes that we implemented modify the basic functionalities of the program, so our project should not be seen as an extension to said application.

Finally, we used the C++ library Voro++ [3] to generate voronoi decompositions, a core aspect of our erosion algorithm.

## Chapter 2

# Previous work and Preliminaries

To understand this project and our contributions to the field, we will first present some background information about the topic as well of previous approaches to our problem (or similar problems), so that the reader can better understand both the problem and our proposed solutions.

## 2.1 Terrains and Landscapes

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### 2.1.1 General weathering effects

### 2.1.2 Freeze-thaw cycle

## 2.2 Terrain Representation and Modeling

Virtual terrains are a fundamental aspect of this thesis, so it is important to review current approaches to represent and work with these types of models. A deep dive into the state-of-the-art on terrain generation can be found on [4], but we are going to present only the most relevant methods for our project, as a full review is out of the scope of this thesis.

### 2.2.1 Terrain Representations

When representing terrains and landscapes there are multiple data structures and representations we can use depending on the ammount of information that we need to store and the nature of the model.

The main model representation techniques can be divided in *Elevation Models*, maninly focused on representing the surface of the terrain, and [Volumetric Models], that aim to capture some interal properties of the terrain as well as more complex structures.

#### Elevation Models

Elevation models are the simplest ways of representing virtual terrains. They consist on assigning an height to every point on a 2D space. In a more formal way, a terrain is represented by a function  $h : \mathbb{R}^2 \rightarrow \mathbb{R}$  that is at least  $C^0$ . From an height function we can easily extract the slope at one point  $s(x, y)$  by using the norm of the gradient:



$$s(x, y) = \|\nabla h(x, y)\| = \left\| \left( \frac{\partial h}{\partial x}, \frac{\partial h}{\partial y} \right) \right\|$$

And from the gradient (projected to  $\mathbb{R}^3$ ) we can easily obtain the normal at one point  $n(x, y)$ :

$$n(x, y) = \frac{\left( -\frac{\partial h}{\partial x}, -\frac{\partial h}{\partial y}, 1 \right)}{\left\| \left( -\frac{\partial h}{\partial x}, -\frac{\partial h}{\partial y}, 1 \right) \right\|}$$

With the height, normal and slope we have all that we need for a basic representation of a terrain. Of course, as we are using a bivariate function to represent the terrain, we can only have one elevation for each point on the domain, and that makes features like caves or overhangs impossible to represent with this method.

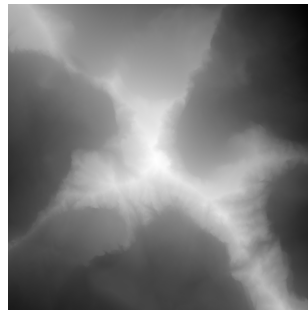
Despite its limitations, elevation models are widely used for its simplicity and memory efficiency as they can capture the general structure of a landscape, given that complex features like overhangs often represent a small part of the overall terrain.

To expand on this topic, elevation models only require a bivariate function to represent the whole terrain, which can capture the whole terrain in a really simple closed-form expression, providing infinite precision with an extremely small ammount of memory. However, constructing functions that represent complex realistic terrains is highly challenging and, for that reason, elevation models commonly use *discrete heightfields* (also called *Digital Elevation Models* or DEM) to represent terrains.

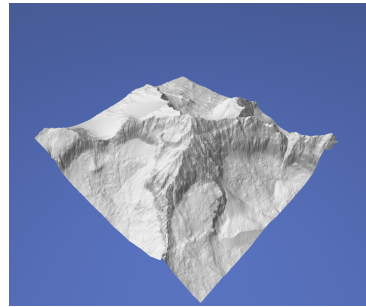
Discrete Heightfields are a collection of altitudes in a 2D grid. In order to create a DEM, you have to divide the domain of the terrain (assuming a rectangular domain) into a grid of  $n \times m$  cells, then, you assign an altitude to every cell of the grid. The grid will only contain concrete values for a set of discrete values, but you can obtain a height for every point of the domain by interpolating multiple values of the heightfield. This method limits the accuracy of the model (as we stop having infinite precision), but it has the advantage of representing arbitrary terrains in a really simple way.

Heightfields are an extremely common way to represent terrains, and they can be easily stored in textures and monochromatic images (as figure 2.1 shows), sometimes making use of the texture interpolation capabilities on the GPU to accelerate computations on heightfields. Specifically, DEMs are used on [2], a project focused on terrain analysis that provides multiple metrics for terrains and whose code has been1 reutilized in this project to implement and test our erosion algorithm.

To finalize the topic of elevation models, **HABLAR SOBRE LAYERED REPRESENTATIONS Y CITAR UN PAPER DE EROSION QUE LAS USE.**



(a) Aneto heightfield



(b) Aneto model on the app

Figure 2.1: Heightfield of the mountain Aneto as a monochromatic image and the terrain produced by said heightfield on our application

## Volumetric Models

### 2.2.2 Terrain Synthesis

## 2.3 Model Decompositions

### 2.3.1 Voxelization and Space Decompositions

### 2.3.2 Tetrahedralization

### 2.3.3 Voronoi Decomposition

## 2.4 Terrain Weathering

### 2.4.1 Physical Models for Erosion

### 2.4.2 Dynamic simulations

## 2.5 Previous Simulator

## Chapter 3

# Model Decomposition

One of the core aspects of simulating erosion on 3D terrains is the model representation that we use. Any erosion simulation applied on a 3D model will modify the model and its properties, so proper data structures that allow for dynamic modifications of the model are key for these types of simulations. Additionally, applying erosion to an object also exposes the *interior* of said object, so any model representation that we use also has to encode some internal information of the model, as internal points of a model might become part of the surface at some point of the process.

In order to represent the internal structure of a model, we decomposed the base model into multiple cells covering the whole volume of the object. A core part of this project has been focused on the model decomposition, as well as on algorithms to produce and improve said decomposition. In this section of the thesis we are going to introduce some basic topics on 3D modeling, explore the potential representations that could be used for terrain erosion and, finally, we are going to take a deep dive into the method we implemented and the improvements we added to increase the quality of the decomposition.

### 3.1 Initial Method

### 3.2 Problems and Solutions

### 3.3 Final Method

### 3.4 Extensions

## Chapter 4

# Erosion

### 4.1 Initial Method

### 4.2 Multi-resolution

### 4.3 Physical Model

### 4.4 Extensions

## Chapter 5

# Implementation

## Chapter 6

# Results

## Chapter 7

# Conclusions

Summary of the previous results, final conclusions and closing thoughts

### 7.1 Future Work

### 7.2 Conclusions

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